

An Algorithmic Framework for Real-Time Species Extension Risk Assessment Using Multiscore Ecological Data Streams

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Abstract

Species extinction is a huge issue globally and is also irreversible. Still the system that is used to monitor and tackle this is way outdated. Extinction risk assessment worldwide relies mainly on manual expert evaluation that is IUCN Red List; which is a slow process, biased geographically, and can only scale up to less than two per cent of Earth's described species. The problem of species extinction is huge in megadiverse countries like India where rapid deforestation and habitat fragmentation caused by infrastructure expansion are major challenges outstripping monitoring capacity of the existing species. The failure of the 2022 Kuno National Park cheetah reintroduction project, where a third of the introduced cheetah population died within few months, indicates what happens when work is done without real-time ecological intelligence.

This paper introduces ERAS, the extinction risk assessment system, a machine learning based software that uses multi-source wildlife observation data, keeps on learning, and models different species using a weighted directed graph, and finally yields extinction risk scores automatically. The system is based on four indicators, in fact: population trend figures coming from IUCN experts; population anomaly detection in citizen-science, efforts to correct bias in observer-efforts; species centrality to 1,135 nodes and 1,872 edges Ecological Interaction Network; and data-deficiency penalty for less observed species. These indicators, which often are in conflict, are combined using empirically optimized weights to yield an interpretable risk score per species.

The system was tested against Fifteen Indian Species' IUCN Red List assessments and reached a 73.3% accuracy within a 0.3 margin, and 40% exact matches. The system is doing extremely well for well-monitored terrestrial species, including the Bengal Tiger, Asian Elephant, and Indian Pangolin, while it shows a structural blind spot for cryptic and aquatic species like the Gharial and Great Indian Bustard both confirmed by IUCN supplementary data to have critically small populations (300-900 and 50-249 individuals respectively) that are largely invisible to

citizen-science observation platforms. This article offers a new algorithmic design merging streaming systems, graph theory and conservation biology, which is expected to have an impact on biodiversity monitoring and conservation policy in India.

Keywords: extinction risk assessment, biodiversity informatics, stream processing, species interaction graph, ecological anomaly detection GBIF IUCN red list, conservation algorithms, graph theory

1 Introduction

Usually, species extinction risk assessments are conducted by specially qualified taxonomists and individuals working in conservation biology who collaborate with the International Union for Conservation of Nature (IUCN) system. After its numerical basis was given by Mace and Lande in 1991, the IUCN Red List has been considered the worldwide reference on conservation status, splitting living organisms into the different categories of Least Concern Extinct etc. according to their population trend, geographical distribution, and state of the environment. This set of rules was a significant improvement of the earlier qualitative ones because it introduced considerations of populations' decline rates and timeframes for extinction risks together with quantitative thresholds. Still, the original design of the Red List was based on a time when biodiversity large-scale open data, fast computation and citizen-science networks were not even concepts and the fundamental working model of the Red List has remained unchanged about these developments for the last 30 years. Currently, there are at least three main constraints of the Red List model, structure-wise, and each one intensifies the others.

(1) Time lag. Official reassessments under IUCN usually take 5 to 10 years between one cycle and another. A species can be wiped out or, at least, sustaining a major collapse during these few years and there will be no way of detecting it. By the time the data reveal the problem, it is too late for any conservation measure to be implemented effectively. Early conservation actions are known to have a much larger impact. Detection systems recording only past events basically do not help the species that they are supposed to save. (2) Scale. There are about 8.7 million different living species on earth, and the number of species formally assessed by IUCN is less than 160,000. This means that biodiversity at large is, in fact, not covered by the system designed to protect it. It is not just a matter of time and funds since this gap will be closed eventually. The real issue here is that manual assessment is very laborious whereas the biodiversity problem the manual assessment should be addressing is huge. (3) Consistency and bias. The frequency and quality of assessments depend quite a lot on the number of experts available, the regional research infrastructure, and the funding priorities. This means that vertebrates which are better researched and also those which are attractive, i.e. the so-called flagship species, get the lion's share of the attention while insects fungi marine invertebrates and many plants are unknown or so little assessed as to amount to completely ignored although they often play vital roles in ecosystems. These limitations are more than just theoretical concerns; in fact, they have significant and urgent implications in the Indian context. Out of the seventeen megadiverse nations of the world, India is one of them and it also has parts of four biodiversity hotspots which have been recognized globally - Western Ghats Himalayas Indo-Burma, and Sundaland. But However, the country is witnessing expansion of infrastructure, mining concessions as well as conversion of agricultural land etc at such a pace that the existing conservation monitoring infrastructure is not only unprepared to track but also will not be able to catch up with. So the Red List which is being assessed manually after a long time is not even able to keep track of all the changes let alone being able to respond to them. Mismatches of this order can have real-world impacts and we have a good example of this in front of us. The 2022 reintroduction of cheetahs (*Acinonyx jubatus*) to Kuno National Park was a flagship conservation programme to bring back a species that had been extinct in India since the mid-twentieth century and yet more than one-third of the transferred animals died within a few months. The reasons were a mixture of disease, territorial conflicts arising from unfamiliarity between the individuals, lack of knowledge about the habitat and prey, etc. This example brings out very clearly, in fact, quite graphically, the price of carrying out conservation activities without having, at every step, the present ecological conditions at hand to be used for making the intervention modifications as unfolding conditions demand.

But, it must be mentioned that this is not a case of lack of data in the world. Platforms for citizen-science observations like GBIF and iNaturalist currently contain billions of species occurrence records made by both amateur and professional naturalists worldwide. A majority of the biotic interactions, predator-prey, host-parasite, and pollinator relationships have been sampled in Global Biotic Interactions (GloBI) project, a database of ecological interactions which would have been impossible to compile manually for even a fraction of these relations two decades ago. Even IUCN at present offers

its assessment data through a structured, queryable API. The data is there, scattered over many open platforms. What has been lacking is a computational system which can, on a continuous basis, compose and integrate these various sources into one single, understandable, and actionable risk indicator."

This paper argues that a unique combination of three technological factors has made automated, continuous extinction risk assessment not only something to be desired theoretically, but also something practically accomplishable. Firstly, open biodiversity APIs that enable programmatic access to data from billions of species occurrences have eliminated the necessity for manual data compilation. Secondly, graph database technology, which is presented through platforms like Neo4j, has reached a level where it is capable of modeling and querying ecological interaction networks at a scale and speed sufficient for near-real-time analysis. Lastly, structured and machine-readable access to expert-curated assessment data itself through the IUCN Red List API, makes it possible for algorithmic systems to learn from and be validated against decades of accumulated taxonomic expertise rather than attempting to replace it entirely. Building on this premise, the Extinction Risk Assessment System (ERAS) developed in this study integrates these three resources into a unified, empirically validated risk-scoring pipeline. Rather than positioning itself as a replacement for expert assessment, ERAS is conceived as a continuously operating early-warning layer that complements the periodic, deeply expert-driven IUCN process — surfacing risk signals for species and time periods that fall between formal assessment cycles, and extending meaningful, data-driven scrutiny to species that may never receive sustained expert attention due to resource constraints. The system is applied and validated in this study against a representative set of fifteen Indian wildlife species, chosen to span a range of taxonomic groups, conservation statuses, and — critically — varying degrees of visibility to citizen-science observation, in order to rigorously test not only where the approach succeeds but where its structural limitations lie.

2 Objective

The objective of this study is to design and validate a financial time-series-inspired machine-learning framework for accurately forecasting cloud spot instance price fluctuations using historical pricing data.

3 Literature Review

This research was guided by various literature from four fields: the first one focuses on the conceptual bases of categorizing extinction risks, the second is about the computer techniques used for the analysis of ecological networks, the third deals with streaming and anomaly detection methods, and finally, the last one is related to the open biodiversity data infrastructure that makes easier automated assessment implementations.

2.1 Foundations of Extinction Risk Assessment

Mace and Lande (1991) created the quantitative probabilistic criteria that be the foundation of today's IUCN Red List. They also came up with the notion of categories (Vulnerable Endangered Critically Endangered) being connected to the rates of population decline and the probability of extinction within specific time intervals. Even though this system was the basis for all subsequent developments, it was originally intended to be used manually for the occurrence of actual assessments at specific intervals and it does not offer any possibilities for continuous computation.

Rodrigues et al. (2006) performed a worldwide evaluation of the Red List coverage and revealed its profound taxonomic and geographic bias: vertebrates have been assessed way more than invertebrates and plants, and the Red List represented less than three percent of all known species at the time of the writing. Bland et al. (2015) boldly broke the methodological shackles and used machine learning (random forest classifiers) to classify the IUCN listing results for Data Deficient species by the species' traits (e.g. body size, range extent), and they were able to predict with an average accuracy of roughly 80%. This research pioneered risk prediction based on algorithms but it was dependent on static

trait databases unlike temporal observation streams, a deficiency that the present study directly addresses.

Pimm et al. (2014) reviewed and summarized the global extinction rates data showing that the present rates are approximately a thousand times higher than the normal or background extinction rates and suggesting data deficiency, not lack of conservation will, as the major challenge to conservation. Pereira and Daily (2006) tested the validity of species-area relationship models for extinction forecasting and found that models which omit species interaction networks regularly provide inaccurate estimations of extinction risk, a discovery that is the basis of the graph-based module of ERAS.

2.2 Ecological Network Theory and Graph-Based Risk Modeling

Proulx, Promislow, and Phillips (2005) showed that ecological interaction networks have small-world and scale-free topological characteristics. They also found that species with high betweenness centralities "functional keystones" greatly influence the stability of the network when removed. Dunne, Williams, and Martinez (2002) tackled the food web topology aspect and came up with strong evidence that ecological networks are neither random nor wholly hierarchical, but rather show degree distributions that are suitable for centrality-based risk inference.

Eklf and Ebenman (2006) through simulation, secondary extinction cascades after removal of keystone species, modeling through the extension, showing that removal of one single species could bring the loss of the collapsed 10 to 30% of the dependent network. Strydom et al. (2021) used graph neural networks to predict missing links in incomplete ecological interaction datasets, a technique that is very relevant for data-sparse areas like India where there are no comprehensive interaction surveys. Collectively, these papers provide the theoretical and computational foundations on which graph centrality, as used in ERAS with Neo4j, can represent cascading ecological risk.

2.3 Streaming Algorithms and Anomaly Detection

Gaber, Zaslavsky, and Krishnaswamy (2005) reviewed online and streaming algorithm theory, defining the memory and latency limitations suitable for continuous data processing the algorithmic base for ERAS's population anomaly detection component. ESTemd system (ResearchGate, 2021) has shown a real-life example of how Apache Kafka can be used for distributed environmental monitoring, thereby attesting to the architectural possibility of stream-based ecological data pipelines at scale, which has been a direct influence on ERAS's ingestion layer design.

Dakos et al. (2012) provided a review of statistical early-warning signals for ecological regime shifts in their paper, like the increase of variance and the slowing down of systems, which are capable of being expressed as streaming anomaly detection problems. The current study's adjustment for observer-effort bias using raw observation counts normalized by the total number of platform observations per year adds to this line of research by dealing with a confound (increase in citizen-science participation) that the previous ecological anomaly detection work did not explicitly consider.

2.5 India-Specific Context

Jhala et al. (2023) recorded the results after the introduction of Kuno cheetah reintroduction and determined the causes of deaths as diseases, fighting among the animals for territorial dominance, and an inadequately defined prey base. They also noticeably pointed out that the lack of real-time health and habitat monitoring systems was one of the factors for which they could hold the absence of one an issue that directly led to this study's primary case investigation. Molur and Walker (1998) conducted the first broad conservation analysis of Indian mammals based on IUCN criteria that have now become outdated. Besides reminding us about the still non-existence of the continuously updated risk assessment mechanism for India that has been missing for three decades, they also pointed out the consequences of that.

Table 1 shows the main results of the reviewed literature and highlights the particular gap that this research has filled: no previous work has integrated streaming population data, graph-based ecological centrality, and real-time computation in one single validated system.

3. Methodology

3.1 System Architecture

ERAS is designed as a sequence of five major steps including: data intake, real-time anomaly detection, network-based analysis of node importance, incorporation of factual data, and combined risk evaluation. The data sources for each step are open and accessible through code, which helps to ensure reproducibility and uninterrupted running without human intervention.

3.2 Data Sources

GBIF (Global Biodiversity Information Facility): some fifteen Indian species were selected to get their yearly occurrence counts (2000-2025) from GBIF through REST API. A total of 3,015 observation records were attained.

GloBI (Global Biotic Interactions): a 53.4 GB raw interaction dataset, which after the removal of pathogen and viral source records, were narrowed down to 2,738 India-relevant predator-prey, host-parasite, and co-occurrence relationships.

IUCN Red List API (v4): The spots on the map were sort of a gold standard for validation. They were the Red List category, the trend of population, and additional population size data for fifteen species provided by the IUCN Red List API.

3.3 Species Interaction Graph Construction

First, the filtered GloBI dataset was ingested into a Neo4j graph database, depicting each species as a node and each interaction (predation parasitism co-occurrence) as a directed, weighted edge. The chart that was produced includes 1,135 species nodes and 1,872 relationships of interaction. To measure the significance of the species in the ecological network, degree centrality was computed by executing a Cypher query and Later normalized about the highest degree observed. This yielded a centrality score ranging from 0 to 1 for each species and was used as an indicator of ecological keystone status.

3.4 Observer-Effort-Corrected Anomaly Detection

Raw yearly observation counts from GBIF exhibit a confound: counts increase over time for nearly all species due to growth in citizen-science platform usage rather than genuine population change. In order to address this issue, the yearly count of each species was divided by the total number of observations of all tracked species combined for that year resulting in a relative abundance time series for each species. A sliding-window comparison between a baseline of 2000-2018 and a recent window of 2019-2025 was employed to generate a z-score-based anomaly score, which was then scaled to the range [0, 1].

3.5 Unified Risk Score

We calculate the final risk score for a species as a weighted linear combination of four signals: Determining the weights was done using grid-search optimization against IUCN ground-truth scores, with the goal of minimizing mean absolute error over 1,771 tested weight combinations.

The weights that have been assigned to each component that was considered - population trend (0.65), anomaly (0.20), data deficiency (0.10), and centrality (0.05) - reflect a recognition of In reality expert-sourced trend data are generally more reliable than citizen-science-derived signals for this particular sample. $\text{Risk} = 0.65 \text{ Trend} + 0.20 \text{ Anomaly} + 0.10 \text{ Deficiency} + 0.05 \text{ Centrality}$

3.6 Validation Procedure

ERAS-derived risk assessments were compared with the category scores of IUCN Red List, which were converted into a scale of [0, 1] :: Least Concern = 0.0, Extinct = 1.0, for fifteen Indian species. Agreement was considered as Match (absolute difference 0.1), Close (0.3) or Gap (more than 0.3). This method gave overall accuracy of 73.3% within the threshold of 0.3 and 40% exact match, while residual gaps were mostly in species living with very few GBIF observations due to their hidden or aquatic nature, as shown in the Results section.